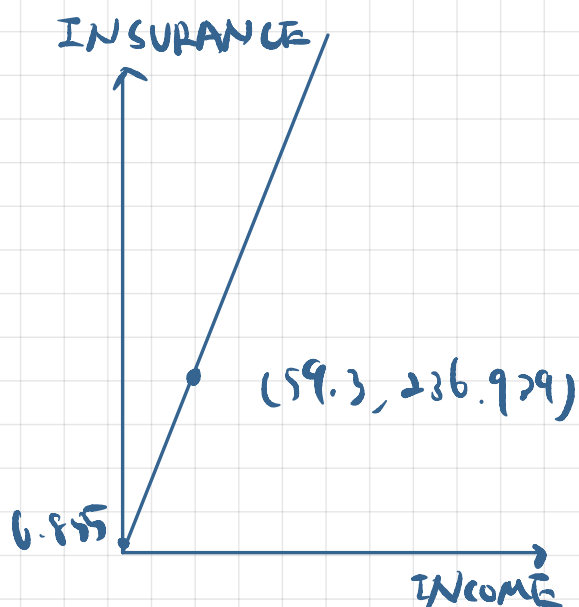


3.18

(a)



$$\begin{aligned}\hat{y} &= 6.855 + 3.880(59.3) \\ &= 6.855 + 230.084 \\ &= 236.939\end{aligned}$$

(b)

INCOME + \$1000 $\rightarrow$ INSURANCE + \$ 2880

$$df = 18, \alpha = 0.05, t_c = 2.101$$

$$CI = b_2 \pm t_c \times se(b_2) = 3.880 \pm 2.101 \times 0.112$$

$$CI = 3.880 \pm 0.2353 \Rightarrow [3.6447, 4.1153]$$

(c)

$$x_n = 100, \hat{y}_n = 6.855 + 3.880(100) = 394.855$$

$$\text{Cov}(b_1, b_2) = -0.746$$

$$\text{Var}(\hat{y}_n) = \text{Var}(b_1) + x_n^2 \text{Var}(b_2) + 2x_n \text{Cov}(b_1, b_2)$$

$$\text{Var}(\hat{y}_n) = (7.383)^2 + (100)^2(0.112)^2 + 2(100)(-0.746)$$

$$\text{Var}(\hat{y}_n) = 54.5057 + 125.44 - 149.2 = 30.7487$$

$$se(\hat{y}_n) = \sqrt{30.7487} \approx 5.545$$

$$99\% (t_{0.005, 18} = 2.878)$$

$$394.855 \pm 2.878 \times 5.545 = 394.855 \pm 15.958 \Rightarrow [378.897, 410.813]$$

(d)

$$H_0: \beta_2 = 5, H_1: \beta_2 \neq 5$$

$$t = \frac{3.880 - 5}{0.112} = \frac{-1.12}{0.112} = -10$$

顯著 5%, $df = 18$, 拒絕 $|t| > 2.101$

0.0 1-101 > 2.101, 強烈拒絕 H_0

(e)

$$H_0: \beta_2 = 1, H_1: \beta_2 > 1$$

$$t = \frac{3.880 - 1}{0.112} = \frac{2.880}{0.112} \approx 25.71$$

顯著 1%, $df = 18$, $t_{0.01, 18} = 2.552$

拒絕: $t > 2.552$

$25.71 > 2.552$, 拒絕 H_0

3.23

$$(a) \text{ PRICE} = \alpha_1 + \alpha_2 \text{ SQFT}^2 + e$$

$$E[\text{PRICE} | \text{SQFT}] = \alpha_1 + \alpha_2 \text{ SQFT}^2$$

$$\frac{dE[\text{PRICE} | \text{SQFT}]}{d \text{SQFT}} = 2\alpha_2 \text{ SQFT}$$

$$\text{when SQFT} = 20, ME = 2 \cdot \alpha_2 \cdot 20 = 40\alpha_2$$

$$H_0: 40\alpha_2 \leq 13; H_a: 40\alpha_2 > 13$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\sigma}_2 - \frac{13}{40}}{se(\hat{\sigma}_2)} \sim t(500-2)$$

$$RR = \{ t \geq t_{0.05498} \approx 1.648 \}$$

$$t_0 = -26.72875$$

$$p\text{-value} = P(t \geq -26.72875 | H_0 \text{ true}) \approx 1$$

$p\text{-value} \approx 1 > 0.05$, 不拒絕 H_0

b) when $SQFT = 40$, $ME = 2 \cdot \alpha_2 \cdot 40 = 80 \alpha_2$

$H_0: 80 \alpha_2 \leq 13$; $H_a: 80 \alpha_2 > 13$

$t = \frac{\hat{\alpha}_2 - \frac{13}{80}}{se(\hat{\alpha}_2)} \sim t(498)$

$t_0 = 4.189467$

$p\text{-value} = 1.65395 \times 10^{-5}$

$p\text{-value} \leq 0.05 \nrightarrow \text{reject } H_0$

c) $E[PRICE | SQFT = 20] = 167.7935$

95% : $[158.0481, 176.6988]$

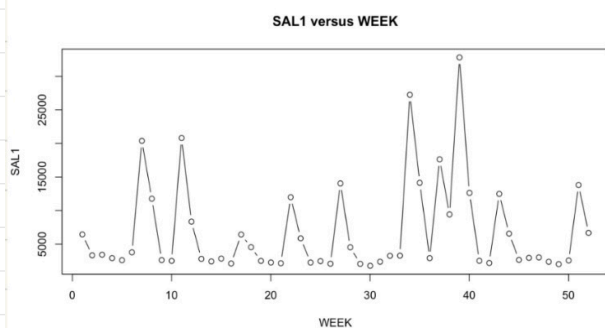
d) $\overline{PRICE | SQFT = 20} = 163.3333$

$163.3333 \in [158.0481, 176.6988]$

Compatible

3.31

a)

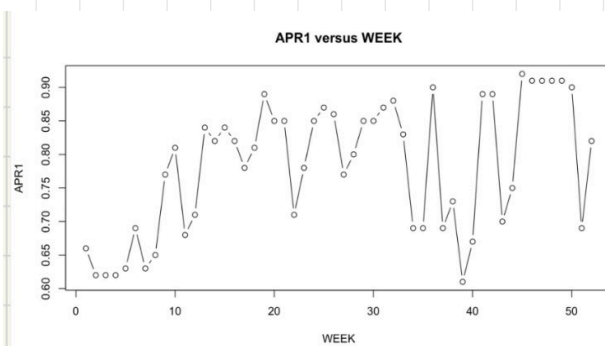


mean : 6719

min : 1762

max : 25820

standard : 6915.083



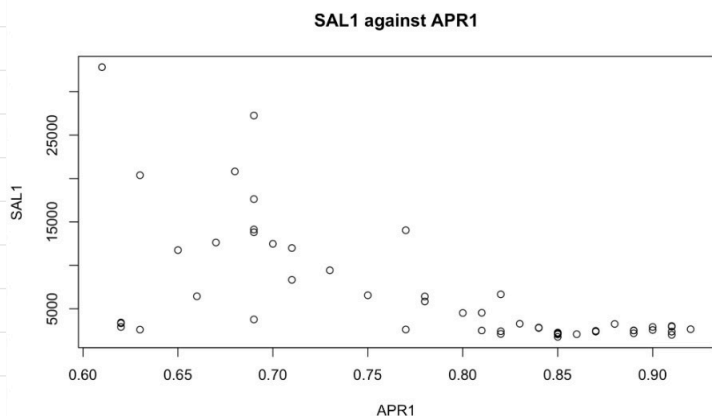
mean : 0.7825

min : 0.6100

max : 0.9200

standard : 0.09803711

(b)



SAL1 and APR1

Inverse relationship

(c) $SAL1 = \beta_1 + \beta_2 PRICE1 + e$

$$\frac{\Delta SAL1}{\Delta PRICE1} = \beta_2$$

$\hat{\beta}_2 = -434.4473$ (代表每增加 1 美分, 銷量平均減少 434)

95% CI : $[-592.2897, -276.049]$

(d) when $PRICE1 = 70$

90% CI : $[8624.934, 11980.87]$

(e) elasticity = $\frac{\Delta SAL1}{\Delta APR1} \cdot \frac{APR1}{SAL1} = \beta_2 \cdot \frac{PRICE1}{SAL1}$

95% : $[-6.898149, -3.21501]$

(f) $H_0: E = -3, H_a: E \neq -3$

$\alpha = 0.10$

$$t = \frac{\hat{E} - (-3)}{SE(\hat{E})} \sim t(52-2)$$

$|t| \geq t_{0.05, 50} = 1.6759$

p-value = 0.02884204

p-value ≤ 0.1 , 拒絕 H_0